

WILL III

Relational Quantum Mechanics

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Abstract

Work in progress...

This paper, the third in the WILL series, applies the principles of Relational Geometry to the quantum realm. From the foundational principle **SPACETIME** $\equiv$ **ENERGY**, we derive the complete structure of the hydrogen atom. By enforcing geometric and topological closure, we demonstrate that quantization is not a separate postulate but an inevitable consequence of a self-consistent relational system. This framework derives the Bohr radius, the quantized energy levels, and reveals the fine structure constant, α , to be nothing more than the kinetic projection parameter, β , of the electron in its ground state. The derivation requires no classical force analogues, probabilistic wavefunctions, or differential equations, establishing atomic structure as a direct manifestation of relational geometry.

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1 Geometric Origin of the de Broglie Relation

We begin with derivation of the de Broglie Relation which build on the results of WILL Part I

Theorem 1.1 (Geometric Origin of the de Broglie Relation). *On the S^1 kinematic manifold of WILL RG, let the horizontal (Amplitude) and vertical (Phase) projections be (β, β_Y) with*

$$\beta_Y \equiv \sqrt{1 - \beta^2} = \frac{1}{\gamma}, \quad E = \frac{E_0}{\beta_Y} = \gamma E_0. \quad \textbf{Inherited from WILL part I}$$

Then momentum and wavelength are necessarily inverse in the sense that

$$p \lambda = (\text{action constant}).$$

Derivation from S^1 Geometry. (1) Momentum from projections. In WILL S^1 kinematics,

$$p = \frac{E}{c} \beta = \frac{E_0}{c} \frac{\beta}{\beta_Y}.$$

(2) Wavelength as inverse projection ratio. The de Broglie wavelength encodes the intrinsic S^1 phase scale. Geometrically,

$$\lambda = \Lambda_0 \frac{\beta_Y}{\beta},$$

for some calibration length Λ_0 that fixes the map from dimensionless phase to physical length.

(3) Product. Multiplying,

$$p \lambda = \left(\frac{E_0}{c} \frac{\beta}{\beta_Y} \right) \left(\Lambda_0 \frac{\beta_Y}{\beta} \right) = \frac{E_0}{c} \Lambda_0.$$

The projections cancel identically, so $p \lambda$ is a constant independent of state.

(4) Calibration to h . Choosing the phase-length calibration so that one full 2π winding on S^1 corresponds to one quantum of action,

$$\frac{E_0}{c} \Lambda_0 = h,$$

we obtain the standard de Broglie relation $\boxed{p \lambda = h}$. □

Remark 1.2 (Calibration and the role of $\hbar$). *The constant $\Lambda_0 = hc/E_0$ is the (reduced) Compton scale associated with the rest invariant E_0 ; this choice implements the identification that a 2π phase winding carries action h . Consequently, $\hbar$ is the calibration invariant translating the dimensionless S^1 phase to SI units of action, and is unchanged under any scale transformation that preserves the winding number.*

2 The Principle of Geometric Closure in Quantum Systems

We continue with the foundational principle established in WILL Part I: any stable, self-contained system must be described by a closed and self-consistent relational geometry.

Interactive visualisation

<https://willrg.com/quantazation.html>

Principle 2.1 (Geometric Closure). *A system in a stable, stationary state must be represented by a geometry that is topologically closed. All relational projections describing the system must return to their initial values after a complete cycle, ensuring the continuity and integrity of the configuration.*

For a bound electron in an atom, this principle has a direct and profound consequence.

Proposition 2.2 (Topological Quantization Condition). *The principle of Geometric Closure, when applied to a bound electron, requires its relational wave to form a standing wave around the orbital circumference. This is expressed by the condition:*

$$n\lambda_n = 2\pi r_n, \quad n = 1, 2, 3, \dots$$

where λ_n is the electron's de Broglie wavelength on the n -th orbital of radius r_n .

Proof. If the wave did not close on itself (i.e., if n were not an integer), destructive interference would prevent the formation of a stable, time-independent configuration. Such a state would be geometrically and energetically inconsistent, and thus is not a permissible solution in WILL Geometry. The integer n is therefore a necessary condition for a stable state's existence. $\square$

Definition 2.3 (The Quantum Number as a Topological Invariant). *The principal quantum number, n , is not an externally imposed parameter but is the **topological winding number** of the energy projection's phase. It counts the number of complete phase rotations required for the geometric configuration to close upon itself.*

Corollary 2.4 (Geometric Quantization of Angular Momentum). *The topological closure condition, combined with derived de Broglie relation $\lambda_n = h/p_n$, directly yields the quantization of angular momentum as a geometric necessity, not a separate physical postulate.*

$$n \frac{h}{p_n} = 2\pi r_n \quad \implies \quad \boxed{p_n r_n = n\hbar}$$

3 The Universal Scale Principle and Relational Projections

As established in Part I, the state of any system is fully described by its potential projection (κ) and kinetic projection (β). The potential projection is governed by a universal principle connecting the system's geometry to its intrinsic limits.

Principle 3.1 (Universal Scale Principle). *The potential projection κ^2 is the dimensionless ratio of a system's intrinsic critical scale to its current observational scale.*

$$\kappa^2 = \frac{r_{\text{critical}}}{r_{\text{current}}}$$

Remark 3.2 (Structural Symmetry of Interactions). *The identical inverse-square form of the gravitational and electromagnetic interactions ($F \propto 1/r^2$) is not a coincidence. It reflects that both are manifestations of the same underlying geometric principle. This compels us to define a critical scale for electromagnetism that is directly analogous to gravity's Schwarzschild radius.*

Definition 3.3 (Electromagnetic Critical Radius). *The electromagnetic critical radius, R_q , is the intrinsic scale at which a particle's electrostatic potential energy budget becomes saturated. In the WILL framework, this corresponds to the radius where the potential energy magnitude equals half the particle's rest energy, consistent with the limit $\kappa = 1$ in our energy budget formalism $|U| = \frac{1}{2}E_0\kappa^2$.*

Definition 3.4 (Electromagnetic Critical Radius). *The electromagnetic critical radius, R_q , is the intrinsic scale at which a particle's electrostatic potential energy budget becomes saturated.*

Because WILL RG removes the ontological separation between "field" and "geometry" (WILL Part A I: Unifying Principle), the saturation condition for any interaction must stem from the same resource limit. The distinction between "gravitational mass" and "electrostatic energy" is an observational artifact, not an ontological one. Therefore, the electrostatic saturation must mirror the gravitational saturation at the Schwarzschild horizon.

Just as the Newtonian gravitational potential at R_s is $|U_g| = \frac{1}{2}mc^2$ (representing the structural limit of the unified resource), we define R_q as the radius where the electrostatic potential energy reaches this same structural limit:

$$|U(R_q)| = \frac{1}{2}E_0$$

Theorem 3.5 (Derivation of the Electromagnetic Critical Radius). *For an electron of mass m_e and charge e , the electromagnetic critical radius is given by:*

$$R_q = \frac{2e^2}{4\pi\epsilon_0 m_e c^2}$$

Proof. We apply the condition from the definition, setting the potential energy magnitude equal to half the rest energy:

$$|U(R_q)| = \frac{1}{2}E_0$$

Substituting the physical expressions for the electron:

$$\frac{e^2}{4\pi\epsilon_0 R_q} = \frac{1}{2}m_e c^2$$

Solving for R_q directly yields the result. This scale is twice the "classical electron radius" and represents the true point of energetic saturation for the electron's potential field in WILL Geometry. $\square$

4 Derivation of Atomic Structure from First Principles

The complete structure of the hydrogen atom is now constrained by a closed system of three geometric principles, with no further assumptions required.

Lemma 4.1 (The Geometric System of Equations). *The stable states of the hydrogen atom are the solutions to the following system:*

1. **Topological Closure:** $\beta_q^2 \approx \frac{n^2 \hbar^2}{m_e^2 c^2 r_n^2}$

2. **Scale Principle:** $\kappa_q^2 = \frac{R_q}{r_n}$

3. **Geometric Closure:** $\kappa_q^2 = 2\beta_q^2$

The non-relativistic approximation for β_q is used, consistent with the low energy scale of the hydrogen atom.

Theorem 4.2 (Derivation of Quantized Atomic Radii). *The allowed orbital radii r_n are derived solely from the simultaneous solution of the geometric system.*

$$r_n = \frac{4\pi\epsilon_0 n^2 \hbar^2}{m_e e^2}$$

Proof. We begin by substituting the expression for β_q^2 from (1) into the geometric closure condition (3):

$$\kappa_q^2 = 2 \left(\frac{n^2 \hbar^2}{m_e^2 c^2 r_n^2} \right)$$

Next, we equate this with the expression for κ_q^2 from the scale principle (2):

$$\frac{R_q}{r_n} = \frac{2n^2 \hbar^2}{m_e^2 c^2 r_n^2}$$

We solve for r_n . Cancelling one factor of r_n from both sides gives:

$$r_n = \frac{2n^2 \hbar^2}{m_e^2 c^2 R_q}$$

Finally, we substitute the derived expression for the critical radius, $R_q = \frac{2e^2}{4\pi\epsilon_0 m_e c^2}$:

$$r_n = \frac{2n^2 \hbar^2}{m_e^2 c^2} \left(\frac{4\pi\epsilon_0 m_e c^2}{2e^2} \right)$$

All dependencies on c and one factor of m_e cancel, yielding the final expression for the quantized radii. $\square$

Corollary 4.3 (The Bohr Radius as a Geometric Consequence). *The length scale of the atom, the Bohr radius a_0 , emerges not from a force balance, but from the intersection of the three geometric principles. For the ground state ($n = 1$):*

$$a_0 = r_1 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2}$$

4.1 Calculation of Energy Levels

The total energy of the electron is the sum of its kinetic and potential energies:

$$E_n = K_n + U_n = \frac{p_n^2}{2m_e} - \frac{e^2}{4\pi\epsilon_0 r_n}. \quad (1)$$

Using the expression for p_n^2 from equation (6):

$$E_n = \frac{1}{2} \frac{m_e e^2}{4\pi\epsilon_0 r_n} - \frac{e^2}{4\pi\epsilon_0 r_n} = -\frac{1}{2} \frac{e^2}{4\pi\epsilon_0 r_n}. \quad (2)$$

Substituting $r_n = n^2 a_0$:

$$E_n = -\frac{1}{2} \frac{e^2}{4\pi\epsilon_0 n^2 a_0}. \quad (3)$$

Since $a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2}$, we have:

$$E_n = -\frac{m_e e^4}{8\epsilon_0^2 n^2 \hbar^2}. \quad (4)$$

This is the standard expression for the energy levels in the hydrogen atom.

$$E_n = -\frac{m_e e^4}{8(2\pi\epsilon_0)^2 n^2 \hbar^2} = 13.6056931396 \text{ eV} \quad (\text{CODATA; match table}). \quad (5)$$

5 The Geometric Origin of the Fine Structure Constant

We now arrive at the one of the central results of this paper. With the atomic length scale a_0 derived strictly from the intersection of geometric limits, we can calculate the electron's relational projections and uncover the identity of a fundamental constant of nature.

Remark 5.1 (Consistency of Geometry). *It is crucial to note that we continue to use the exact same RG principles established in WILL Parts I and II:*

- **Kinematic Carrier** (S^1): Enforces closure via $\beta^2 + \beta_Y^2 = 1$.
- **Potential Carrier** (S^2): Enforces closure via $\kappa_X^2 + \kappa^2 = 1$.

We strictly reject any ad-hoc introduction of hyperbolic or open geometries. Both cosmological and atomic systems are governed by the same compact, circular invariants.

Theorem 5.2 (The Fine Structure Constant as the Ground-State Kinetic Projection). *The fine structure constant, α , is identical to the kinetic projection, β_1 , of the electron in the ground state of the hydrogen atom.*

$$\boxed{\alpha \equiv \beta_1}$$

Proof. We use the established geometric system to find the value of β_1 .

1. From the **Geometric Closure Condition** (Principle 2.1), the balance between carriers dictates:

$$\beta_1^2 = \frac{1}{2} \kappa_1^2$$

2. From the **Scale Principle** (Principle 3.1), the potential projection is the ratio of the critical scale to the orbital radius:

$$\kappa_1^2 = \frac{R_q}{a_0}$$

Combining these with [closure theorem](#) yields the projection equation:

$$\beta_1^2 = \frac{1}{2} \frac{R_q}{a_0}$$

Substituting the derived expressions for the electromagnetic critical radius R_q and the Bohr radius a_0 :

$$\beta_1^2 = \frac{1}{2} \left(\frac{2e^2}{4\pi\epsilon_0 m_e c^2} \right) \left(\frac{m_e e^2}{4\pi\epsilon_0 \hbar^2} \right)$$

Simplifying the terms:

$$\beta_1^2 = \frac{e^4}{(4\pi\epsilon_0)^2 \hbar^2 c^2} = \left(\frac{e^2}{4\pi\epsilon_0 \hbar c} \right)^2$$

We recognize the term in parentheses as the standard definition of the fine structure constant, α .

$$\beta_1^2 = \alpha^2 \implies \beta_1 = \alpha$$

The theorem is proven. □

Remark: A Derived Constant

The fine structure constant has historically been regarded as an empirical, fundamental parameter ($\approx 1/137$). This derivation might highlight its true ontological identity: as the unique kinematic projection compatible with geometric closure of a charged fermion.

6 Emergence of Quantized Energy Levels

The final piece of the atomic puzzle, the energy spectrum, now follows as a direct consequence of the derived geometry.

Theorem 6.1 (Quantized Energy Levels). *The energy levels of the hydrogen atom are given by:*

$$E_n = -\frac{\alpha^2 m_e c^2}{2n^2}$$

Proof. The total energy E_n of a stable bound state is given by the virial relation, which in our framework is a consequence of the geometric closure condition $\kappa_q^2 = 2\beta_q^2$. This implies $E_n = -K_n$, where the kinetic energy budget is $K_n = \frac{1}{2} m_e c^2 \beta_q^2$. Thus:

$$E_n = -\frac{1}{2} m_e c^2 \beta_q^2$$

For the ground state, we established $\beta_1 = \alpha$. The quantization condition implies that the momentum scales inversely with the principal winding number: $p_n \propto 1/n$. Since $\beta_q \propto p_n$, the kinetic projection scales as:

$$\beta_q = \frac{\beta_1}{n} = \frac{\alpha}{n}$$

Substituting this into the energy expression yields the final result:

$$E_n = -\frac{1}{2}m_e c^2 \left(\frac{\alpha}{n}\right)^2 = -\frac{\alpha^2 m_e c^2}{2n^2}$$

□

7 Spectral Lines and Rydberg Formula

The emitted photon's energy is:

$$E_{\text{photon}} = E_{n_i} - E_{n_f}. \quad (6)$$

Using Planck's relation:

$$hf = E_{\text{photon}}, \quad (7)$$

we derive the spectral formula:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right). \quad (8)$$

7.1 Numerical Results and Comparison

Transition	Computed λ (nm)	Experimental λ (nm)
$3 \rightarrow 2$	656.34	656.3
$4 \rightarrow 2$	486.17	486.1
$5 \rightarrow 2$	434.08	434.0
$6 \rightarrow 2$	410.21	410.2

Table 1: Computed and experimental spectral lines.

8 The Unified State as a Geometric Equilibrium

Our previous investigations revealed that non-relativistic models and simple relativistic corrections fail to describe the fine structure of high-Z atoms. This failure proves that a perturbative approach is insufficient. The complete atomic state is not a simple sum of parts, but a single, unified geometric object.

Principle 8.1 (The Quantum State as Geometric Equilibrium). *A stationary quantum state does not evolve in time. It is a point of stable geometric equilibrium in the composite phase space describing the system. Its observable properties, including energy, are determined strictly by the coordinates of this equilibrium point on the relational carriers.*

RG Method: Standard approaches often invoke hyperbolic geometry (boosts) to describe relativistic effects. However, in WILL, the system is closed. Therefore, the geometry of the phase space determining the "fine structure" must remain consistent with the **circular invariants** of S^1 and S^2 . We seek a solution where the interaction term acts as a geometric projection within a closed Pythagorean system, not as a hyperbolic deformation.

9 The Relational Geometry of Angular Phase Space

The fine structure splitting arises from the interaction between the primary (orbital) and orthogonal (spin) modes. The strength of this interaction is quantified by the coupling parameter $\beta_q = Z\alpha/n$. In the weak-field limit ($Z\alpha \rightarrow 0$), the angular quantum numbers can be treated as "flat," meaning they are not affected by the relativistic phase shift. However, in the strong-field regime, the geometry deforms.

Proposition 9.1 (Interaction-Induced Phase Shift). *The kinetic interaction, quantified by the projection β_q , induces a phase shift in the angular quantum numbers via the Closure Constraint ($\beta_q^2 + \beta_{Yq}^2 = 1$). Consequently, the integer quantum numbers must be corrected to account for this geometric contraction.*

Theorem 9.2 (The Pythagorean Partition of Phase Space). *The geometry of all relational projections must obey the constraint of their respective relational carrier (S^1). Unlike standard relativity which often invokes hyperbolic geometry for kinematics, WILL Relational Geometry imposes **circular closure**. The total topological capacity of a state is distributed between its interaction component and its structural component according to the Pythagorean invariant $C^2 = A^2 + B^2$.*

Proof. This is a direct application of the universal structure of Relational Geometry established in "WILL RG Part I". Since the carrier S^1 is compact and closed, any projection β implies a complementary projection β_Y such that their squares sum to unity. When applied to angular momentum, the total integer capacity (k) acts as the invariant radius (hypotenuse). The interaction strength ($Z\alpha$) acts as one projection (cathetus). Therefore, the remaining structural component must form the second cathetus of a right triangle in phase space. $\square$

This theorem allows us to define the true, relativistic angular number.

Definition 9.3 (Relativistic Angular Number). *In the closed phase space:*

- The integer angular number $k = j + 1/2$ represents the **Hypotenuse** (Total topological capacity).
- The interaction coupling $Z\alpha$ represents the **Interaction Projection** (The portion of capacity consumed by the field).
- The resulting invariant k' represents the **Structural Projection** (The effective angular momentum).

Corollary 9.4 (Derivation of k'). *From the Pythagorean geometry of the phase space ($\text{Hypotenuse}^2 = \text{SideA}^2 + \text{SideB}^2$), the relativistic angular number k' is derived as:*

$$k^2 = (Z\alpha)^2 + (k')^2$$

Solving for the structural component:

$$k' = \sqrt{k^2 - (Z\alpha)^2}$$

9.1 Quantized Relational Geometry

To maintain full structural symmetry between the classical and quantum domains, we extend the relational invariants (κ, β) into their quantized counterparts (κ_q, β_q) .

Definition. The quantized projections obey the same closure (??) relations as their classical analogues:

$$\boxed{\kappa_q^2 = \frac{R_q}{r_n} \quad \text{and} \quad \kappa_q^2 = 2\beta_q^2}$$

where:

- R_q is the **critical quantum radius**, the electromagnetic scale.
- r_n is the quantized orbital radius corresponding to the principal winding number n .
- β_q represents the kinetic projection of the quantum state.

Relational Consistency

The same relational law applies at all scales:

$$\boxed{\kappa^2 = 2\beta^2} \longrightarrow \boxed{\kappa_q^2 = 2\beta_q^2}$$

Quantization is the discretization of geometric closure, not the introduction of a separate quantum postulate. Thus α ceases to be a 'number about electromagnetism' and becomes a geometric ratio defining the phase structure of all closed energy systems.

9.2 The Unified Quantum Number and Total Energy

A stable state is defined by its complete set of winding numbers. The energy is determined by a unified quantum number that combines the radial and angular components within the "curved" geometry.

Definition 9.5 (Unified Quantum Number). *The unified quantum number, n_{eff} , which holistically describes the geometric state, is the linear sum of the radial segment and the projected angular segment:*

$$n_{eff} = n_r + k'$$

Given that $n_r = n - k$ (where n is the principal quantum number), this becomes:

$$n_{eff} = (n - k) + \sqrt{k^2 - (Z\alpha)^2}$$

The total energy is a fundamental projection based on this single, unified quantum number.

Theorem 9.6 (The Unified Relativistic Energy Formula). *The total relativistic energy of an electron is a projection of its rest energy, $m_e c^2$, determined by the ratio of the interaction strength to the total effective geometry. The formula is:*

$$\boxed{E_{n,j} = \frac{m_e c^2}{\sqrt{1 + \left(\frac{Z\alpha}{n_{eff}}\right)^2}}}$$

Proof. This form arises directly from the projection on S^1 . If we construct a phase triangle where the "Interaction" is $Z\alpha$ and the "Structure" is n_{eff} , the kinetic angle θ is defined by $\tan \theta = \frac{Z\alpha}{n_{eff}}$. The total energy corresponds to the cosine projection of the rest energy (the "time-like" component on S^1):

$$E = E_0 \cos \theta = E_0 \frac{1}{\sqrt{1 + \tan^2 \theta}} = \frac{E_0}{\sqrt{1 + \left(\frac{Z\alpha}{n_{eff}}\right)^2}}$$

Substituting the expanded form of n_{eff} yields the exact Sommerfeld-Dirac fine structure formula. $\square$

9.3 Comparison with the Dirac Equation Approach

The formula we have derived is numerically identical to the well-known Sommerfeld fine-structure formula, which gives the exact energy levels of the stationary states of the Dirac equation. It is therefore crucial to clarify the profound difference in methodology.

Remark 9.7 (On the Divergence of Methods). *To arrive at this result, the Dirac approach begins by postulating a relativistic wave equation, synthesizing SR and QM through abstract algebraic objects (spinors and gamma matrices). The WILL approach, in contrast, is generative. It begins with a single geometric principle ($SPACETIME \equiv ENERGY$) and derives the energy formula as a necessary consequence of **Circular Geometric Closure**.*

Methodological Comparison	
Dirac Equation Approach	WILL Relational Geometry Approach
Starts with synthesis of SR and QM postulates.	Starts with a single principle of geometric unity.
Introduces abstract spinors and matrices.	Derives quantities from Pythagorean phase projections.
Solves differential equations for eigenvalues.	Solves for closed geometric triangles.
Nature: Descriptive. Models reality via operators.	Nature: Generative. Derives reality from necessity.

Corollary 9.8. *The Dirac equation is a successful operator-based representation of the underlying relativistic geometry of a quantum state. Our derivation does not replace it, but rather hints at its ontological foundation.*

10 The Geometric Origin of Spin and Exclusion

Having established the geometry of a single electron, we must now address the rules governing multi-electron systems. In standard quantum mechanics, the "spin" of a particle and the Pauli Exclusion Principle are introduced as independent axioms. In WILL Relational Geometry, these properties must emerge as topological necessities of the closed carriers S^1 and S^2 .

10.1 Spin as Topological Chirality

We have treated the phase projection β_Y (associated with the S^1 carrier) as a scalar magnitude representing the "internal clock" or time-like component of the state. However, a closed loop on a manifold S^1 possesses a topological property: **direction of winding**.

Definition 10.1 (Topological Chirality). *For any closed energy flow on S^1 , there exist exactly two distinct, non-deformable winding directions relative to the surface normal of the carrier S^2 :*

1. **Right-Handed Flow** ($\odot$): *Corresponds to positive helicity (Spin "Up", $+1/2$).*
2. **Left-Handed Flow** ($\ominus$): *Corresponds to negative helicity (Spin "Down", $-1/2$).*

These two states are energetically identical ($\delta_+ = \delta_-$) but topologically distinct. One cannot be transformed into the other without breaking the closure of the loop.

Theorem 10.2 (The Origin of Spin). *"Spin" is not an intrinsic angular momentum of a point particle. It is the **Topological Chirality** of the standing wave's phase integration. The value $1/2$ reflects the geometric fact that the phase loop S^1 is coupled to the spherical surface S^2 (Hopf fibration topology), requiring 720° (two full rotations) to return the coupled system to its original state.*

10.2 The Principle of Topological Exclusion (Pauli)

The exclusion principle, which prevents matter from collapsing, is traditionally postulated as the antisymmetry of the wavefunction. In our generative framework, it is a consequence of the *Identity of Indiscernibles* applied to geometric resonances.

[The Principle of Unique Address] In a relational universe, location is defined by relations. If two entities possess identical sets of relational projections (same Energy δ , same Scale n , same Topology k), they are not two entities; they are a single entity with double amplitude.

Theorem 10.3 (Geometric Exclusion). *To exist as distinct, co-located entities within the same potential well (atom), two energy flows must differ by at least one geometric invariant. Since the spatial invariants (n, k, m) define the "shape" of the orbital, the only remaining degree of freedom is the **Chirality** of the phase flow.*

Corollary 10.4 (Capacity of a Geometric State). *A single spatial geometric state (defined by n, l, m) can host exactly two energy flows:*

- *One flow with Right-Handed Chirality ($\odot$).*
- *One flow with Left-Handed Chirality ($\ominus$).*

Any attempt to add a third flow implies duplicating a chirality, causing the flows to merge (constructive interference) and forcing the system to jump to a higher energy scale ($n+1$) to maintain closure. This is the geometric derivation of the Pauli Exclusion Principle.

10.3 The Foundation of Chemical Structure

This exclusion mechanism is the origin of valence and chemical bonding.

- **Open Shell:** An orbital with only one chirality ($\odot$) is geometrically unbalanced. It seeks a partner ($\oslash$) to close the "phase current."
- **Chemical Bond:** When two atoms share electrons, they are simply satisfying the topological requirement to pair opposite chiralities within a shared geometric envelope.
- **Closed Shell:** An orbital with both chiralities ($\odot + \oslash$) is a topologically neutral, self-satisfied structure (Noble Gas configuration).

Thus, chemistry is the mechanics of interlocking geometric chiralities.

11 The Schrödinger Equation as the Result of Ontological Collapse

The Schrödinger equation arises as the algebraic shadow of the second ontological collapse—the fusion of two amplitude–phase manifolds (β, κ) into a single complex energetic amplitude ψ .

I. Pre–Collapse Structure

Before collapse, the system is described by two real projections of one conserved energetic resource:

$$E_0 \Rightarrow (\beta, \kappa), \quad \text{with invariant} \quad \boxed{\kappa^2 = 2\beta^2}$$

Here β denotes the kinematic projection (electromagnetic channel) and κ the potential projection (gravitational channel). Each belongs to its own relational circle:

$$(\beta_X, \beta_Y) \in S^1, \quad (\kappa_X, \kappa_Y) \in S^1,$$

parameterized as

$$\beta = \beta_X = \cos \theta_1, \quad \kappa = \kappa_Y = \sin \theta_2,$$

linked by the geometric coupling

$$\boxed{\sin^2 \theta_2 = 2 \cos^2 \theta_1} \quad \Leftrightarrow \quad \boxed{\kappa^2 = 2\beta^2}.$$

Lemma 11.1 (Dual–Projection Closure). *A closed relational configuration maintains constant energetic balance between both circles:*

$$\delta(\kappa^2 - 2\beta^2) = 0.$$

Proof. Differentiation gives $2\kappa \delta\kappa - 4\beta \delta\beta = 0$, expressing compensatory exchange between potential and kinematic projections. $\square$

II. The Second Ontological Collapse

Definition 11.2 (Collapse of Dual Projections). *The **second ontological collapse** identifies the two projection circles under a single complex representation:*

$$(\beta, \kappa) \longrightarrow \psi \equiv \beta + J \kappa, \quad J^2 = -1.$$

The operator J is the generator of orthogonal rotation between the energetic projections.

This identification fuses the potential and kinematic amplitudes into one entity, so that their relational rotation becomes a phase factor of ψ . What was originally a two-point exchange now manifests as a single complex oscillation.

Lemma 11.3 (Relational Rotation Rule). *A minimal closed exchange of energy between (β, κ) is represented by a rotation of phase α :*

$$\psi' = e^{J\alpha}\psi, \quad \alpha = \frac{\Delta S}{\hbar},$$

where ΔS is the incremental action associated with the exchange.

Proof. The transformation preserves $|\psi|^2 = \beta^2 + \kappa^2$; the invariant $\kappa^2 = 2\beta^2$ constrains the rotation ratio. The exponential expresses the algebraic form of this relational rotation. $\square$

III. Discrete Closure and the Spectral Equation

Theorem 11.4 (Quantization from Discrete Closure). *If after N discrete rotations the system returns to its initial configuration,*

$$\psi_N = \psi_0,$$

then the accumulated phase satisfies

$$N \alpha = 2\pi n, \quad n \in \mathbb{Z}.$$

With $\alpha = \Delta S/\hbar$, discrete closure enforces quantized action and energy levels.

Proof. Closure requires the total rotation to equal an integer multiple of a full circle: $N\Delta S = 2\pi n\hbar$. The energy eigenvalues follow directly from $\mathcal{E}_n = \frac{\partial S_n}{\partial t}$. $\square$

Corollary 11.5 (Spectral Equation). *Expanding the discrete rotation to second order in $\alpha \ll 1$ gives*

$$\psi_{k+1} \approx \psi_k + J \frac{\mathcal{E}}{\hbar} \psi_k - \frac{1}{2} \left(\frac{\mathcal{E}}{\hbar} \right)^2 \psi_k.$$

Summing over neighboring sites along a closed contour produces the discrete Laplacian $\Delta_{disc}\psi_j = \psi_{j+1} - 2\psi_j + \psi_{j-1}$, leading to

$$\left(-\frac{\hbar^2}{2m} \Delta_{disc} + V \right) \psi = \mathcal{E} \psi,$$

where m and V are the inertial and potential calibrations defined by $E_0 = mc^2$ and $V = \frac{E_0}{2}(\kappa^2 - \kappa_^2)$. In the continuum limit $\Delta_{disc} \rightarrow \nabla^2$, the stationary Schrödinger equation emerges.*

IV. Interpretation

Interpretive Summary

- The Schrödinger equation represents the stationary condition of discrete relational closure.
- The complex unit i is the algebraic signature of the fused duality between the β and κ manifolds.
- Quantization follows from finite winding on the relational manifold.

Quantum Mechanics arises from the discrete closure of a collapsed two-projection geometry.

Thus, the so-called “mystery of the quantum” is a residue of two successive ontological collapses — one collapsing relational space into position, the other collapsing relational energy into ψ .

12 Geometric Origin of the Uncertainty Principle

In the WILL framework, the uncertainty principle is not a postulate of quantum theory, nor a consequence of non-commuting operators. It is a direct geometric necessity arising from the closure of energy projection on the unit circle S^1 . This section derives the principle from first principles, without invoking wavefunctions, probabilities, or external measurement axioms.

12.1 Phase Winding as the Ontological Basis of Quantization

The quantum number n is not an externally imposed label but the **topological winding number** of the electron’s energy projection around its orbital path. This follows directly from the foundational principle:

$$\boxed{\text{SPACETIME} \equiv \text{ENERGY}}$$

For a closed orbit of radius r_n , the projection must return to its initial phase after one full circuit. This enforces the geometric closure condition:

$$n\lambda_n = 2\pi r_n, \quad n = 1, 2, 3, \dots \quad (9)$$

where λ_n is the de Broglie wavelength. The integer n counts the number of complete phase rotations - i.e., the number of times the projection wraps around S^1 before closing. No state with $n = 0$ exists; such a configuration would correspond to the absence of a closed projection, and thus to no physical electron.

12.2 Minimal Phase Grain and Observational Granularity

Because the phase must close after n windings, the smallest physically distinguishable phase increment is bounded by:

$$\Delta\theta \geq \frac{2\pi}{n}. \quad (10)$$

This is not a limitation of instrumentation but a **limit of definability**: any phase difference smaller than $2\pi/n$ cannot be distinguished from a full winding and therefore lacks relational identity. The quantity $2\pi/n$ is the *minimal phase grain* - a topological invariant of the closed geometry.

12.3 Orthogonal Projections on S^1 and Their Uncertainties

The unit circle S^1 supports two orthogonal relational projections:

- **Amplitude Component** (β_X): $\beta_X = \cos \theta$, representing the relational (kinetic) aspect of the projection. This manifests physically as momentum.
- **Phase Component** (β_Y): $\beta_Y = \sin \theta$, representing the internal (temporal) aspect. This governs proper time and frequency scales.

For small variations, the uncertainties in these projections are:

$$\Delta\beta_X = |\sin \theta| \Delta\theta, \quad (11)$$

$$\Delta\beta_Y = |\cos \theta| \Delta\theta. \quad (12)$$

Their product is therefore:

$$\Delta\beta_X \Delta\beta_Y = |\sin \theta \cos \theta| (\Delta\theta)^2 = \frac{1}{2} |\sin 2\theta| (\Delta\theta)^2. \quad (13)$$

Substituting the minimal phase grain $\Delta\theta \geq 2\pi/n$, we obtain the **Geometric Uncertainty Relation**:

$$\Delta\beta_X \Delta\beta_Y \geq \frac{1}{2} |\sin 2\theta| \left(\frac{2\pi}{n} \right)^2 \quad (14)$$

This relation is purely topological and algebraic. It requires no probabilistic interpretation, no Hilbert space, and no external observer. It is the inevitable consequence of demanding closure on a finite, self-contained projection.

Epistemic Disclaimer

Equation (12.3) is not an approximation or empirical rule. It is a necessary condition for the existence of a well-defined, closed energy projection. Any attempt to interpret β_X and β_Y as independent classical variables will violate this bound and produce unphysical states.

12.4 Calibration to Physical Observables: Emergence of Planck's Constant

The geometric uncertainty relation (Eq. (12.3)) is expressed in dimensionless projection space. To connect it with empirical physics, we must map these projections to measurable quantities. This mapping arises directly from the ontological nature of a particle in WILL Geometry.

The Ontological Origin of Spatial Uncertainty (Δx). In the WILL framework, a particle is not a point object existing in spacetime; it is a quantized unit of spacetime itself, whose minimal spatial extent is its de Broglie wavelength λ_n . This is a direct consequence of the principle $\text{SPACETIME} \equiv \text{ENERGY}$. The "position" of such a wave-like quantum is therefore inherently delocalized.

Consequently, the fundamental, irreducible uncertainty in its position, Δx , is not a limit of measurement but a **definition of the particle's own geometric scale**:

$$\Delta x \approx \lambda_n. \quad (15)$$

Momentum and the Emergence of the Heisenberg Relation. The uncertainty in momentum, Δp , remains the scaled amplitude projection as previously defined:

$$\Delta p = m_e c \Delta \beta_X. \quad (16)$$

The de Broglie relation, $p_n = h/\lambda_n$, can be rewritten as $p_n \lambda_n = h$. By combining this with our geometric definition of Δx from Eq. (12.4), the Heisenberg Uncertainty Principle emerges not as a postulate, but as an algebraic identity:

$$\boxed{\Delta x \cdot \Delta p \approx h}. \quad (17)$$

This relation is thus shown to be a direct consequence of the particle's ontological status as a self-contained quantum of spacetime. This physical uncertainty is the manifest expression of the underlying Geometric Uncertainty Relation (Eq. (12.3)), with the two being linked via the mappings $\Delta p \propto \Delta \beta_X$ and $\Delta x \propto \lambda \propto \Delta \beta_Y$.

Cycle-averaged geometric bound. Over one closed winding on S^1 , the average of $|\sin 2\theta|$ equals $2/\pi$. Hence the cycle-averaged product obeys

$$\left\langle \Delta \beta_X \Delta \beta_Y \right\rangle_{\text{cycle}} \geq \frac{1}{\pi} (\Delta \theta)^2 \geq \frac{1}{\pi} \left(\frac{2\pi}{n} \right)^2 = \frac{4\pi}{n^2}.$$

This bound is purely geometric (closure on S^1) and refers to a full winding, not to a pointwise angle.

The Nature of Planck's Constant. Critically, $\hbar$ is not a fundamental constant of nature but a **conversion factor** between topological winding and physical scale. It is the calibration invariant under any scale transformation that preserves the winding number. It translates dimensionless geometry of phase into the dimensional world of meters, kilograms, and seconds.

Epistemic Note

In WILL RG, $\hbar$ is epiphenomenal. The fundamental invariant is the winding number n ; $\hbar$ is a human artifact for translating topological closure into SI units.

12.5 Superposition and Collapse as Phase Coherence and Locking

In standard quantum mechanics, superposition is a linear combination of states, and collapse is an ill-defined measurement-induced transition. In WILL Geometry, both are reinterpreted geometrically.

Superposition = Unresolved Winding. A system exhibits interference when its phase projection remains **coherent across multiple paths**. For example, in the double-slit experiment, the electron’s phase is not yet locked to any external reference; its winding number is defined only relative to the global experimental setup. Multiple geometric trajectories remain compatible, and their phase differences produce interference.

Collapse = Phase Locking via Energy Symmetry. When the system interacts with an environment (detector, air molecule, thermal photon), the Energy Symmetry Law enforces:

$$\Delta E_{\text{system} \rightarrow \text{env}} + \Delta E_{\text{env} \rightarrow \text{system}} = 0. \quad (18)$$

This mutual closure requires the system to adopt a **definite phase configuration** relative to the environment. The winding network locks, eliminating compatibility between multiple paths. Interference vanishes - not because of “observation,” but because of **energetic accountability**.

No wavefunction collapse postulate is needed. No observer consciousness. Only geometric closure.

Key Insight

Decoherence is not probabilistic - it is the deterministic resolution of phase ambiguity under the constraint of global energy symmetry.

12.6 Path Integral as Sum Over Winding Configurations

Feynman’s path integral formalism posits that a particle “samples” all possible trajectories, weighted by $e^{iS/\hbar}$. In WILL Geometry, this is reinterpreted without integrals or complex amplitudes.

The physical process is a **sum over all phase winding configurations** that satisfy global closure. Each configuration corresponds to a distinct way the projection can wrap around its geometric manifold while maintaining consistency with boundary conditions.

- The “classical path” is the configuration with **stationary phase** - i.e., minimal phase shear across the network.
- Constructive interference occurs when phase differences between paths equal $2\pi m$ (integer winding match).
- Destructive interference occurs for half-integer mismatches.

No complex numbers are required. The “amplitude” is simply the **geometric compatibility** of a winding configuration with the global closure constraint. The path integral is thus reduced to a combinatorial count of topologically allowed states.

This reframing eliminates the ontological baggage of infinite-dimensional configuration spaces and restores the path integral to its geometric essence: a catalog of self-consistent phase windings.

Summary

- $\hbar$ emerges from phase granularity, not as a fundamental constant.
- Superposition = unresolved winding; collapse = phase locking via energy symmetry.
- Path integral = sum over topologically allowed winding configurations.
- The exponential factor $e^{iS/\hbar}$ of Feynman's formulation is the algebraic shorthand for counting phase rotations of magnitude $S/\hbar = 2\pi n$.

All quantum phenomena arise from the geometry of closed projections - no additional postulates required.

12.7 Entanglement as Shared Phase Origin

In standard quantum mechanics, entanglement is a non-local correlation with no classical analogue. In WILL Geometry, it arises naturally from the **topological unity of phase winding**.

Consider two electrons created in a common process (e.g., atomic decay or pair production). Their energy projections originate from a **single closed phase network** on a shared manifold. This means their winding numbers are not independent but **co-defined** by a global closure condition.

Let the total phase of the composite system be:

$$\Phi_{\text{total}} = \phi_A + \phi_B = 2\pi n_{\text{total}},$$

where n_{total} is a fixed integer. Any local change in ϕ_A must be compensated by ϕ_B to preserve closure:

$$\delta\phi_A + \delta\phi_B = 0.$$

This is not “spooky action at a distance.” It is **relational bookkeeping**: the two subsystems are parts of a single geometric whole. Measuring one fixes its phase relative to the environment, which **locks the other** to maintain global energy symmetry:

$$\Delta E_{A \rightarrow \text{env}} + \Delta E_{\text{env} \rightarrow A} = 0 \quad \Rightarrow \quad \text{phase of } B \text{ adjusts accordingly.}$$

Bell's theorem is not violated because the correlations were **never separable** - they were born from a common phase origin. No hidden variables are needed; only **topological unity**.

Key Insight

Entanglement is not a property of particles - it is a property of the **phase network** that defines them.

12.8 Quantum Field Theory as a Network of Interlocked Circles

In WILL Geometry, a quantum field is not a fundamental entity but a **dense network of interlocked S^1 projections**, each representing a local degree of freedom of WILL=spacetime-energy.

Particles as Localized Winding Excitations. A particle corresponds to a **localized defect** in the phase network - a region where the winding number differs from the ground state. Creation and annihilation are not operator actions but **topological transitions**: adding or removing a winding unit.

Vacuum as Ground-State Winding. The vacuum is not “empty.” It is the **lowest-energy winding configuration** consistent with global closure. Virtual particles are temporary **phase fluctuations** that do not violate energy symmetry because they are confined within the uncertainty bound:

$$\Delta E \Delta t \geq \hbar.$$

Gauge Symmetries as Winding Conservation Laws.

- - **U(1) symmetry (electromagnetism)**: conservation of total phase winding on S^1 .
- - **SU(2) symmetry (weak force)**: conservation of winding on a 3-sphere S^3 (double cover of rotation group).
- - **SU(3) symmetry (strong force)**: conservation of winding on a higher-dimensional manifold with 8 topological generators.

Renormalization is not a fix for infinities but a **rescaling of the winding granularity** to match observational resolution. It corresponds to a change in resolution of the phase network, not to mathematical regularization.

No Lagrangian density is needed. The dynamics of the field are encoded in the **algebraic closure** of the network:

$$\sum_{\text{nodes}} \Delta n_i = 0,$$

ensuring global phase consistency.

Epistemic Note

Quantum field theory in WILL is not a theory of fields in spacetime - it is a **theory of spacetime as a field of phase relations**.

12.9 Final Synthesis: The WILL Uncertainty Principle

We now unify all quantum phenomena under a single geometric principle.

Definition 12.1 (WILL Uncertainty Principle). *On any closed manifold encoding a physical observable, the product of uncertainties between two non-parallel projections is bounded below by the square of the minimal phase grain, set by the topological winding number:*

$$\Delta A \Delta B \geq |G(A, B)| \left(\frac{2\pi}{n} \right)^2$$

where $G(A, B)$ is the geometric coupling (e.g., $\frac{1}{2} |\sin 2\theta|$ for S^1), and n is the winding number of the closed projection.

This principle explains:

- **Position-momentum uncertainty:** $A = x, B = p \rightarrow$ projections on S^1 .
- **Energy-time uncertainty:** $A = E, B = t \rightarrow$ conjugate phase and frequency.
- **Angular momentum uncertainty:** $A = L_x, B = L_y \rightarrow$ non-orthogonal projections on S^2 .

The “commutator” $[A, B] = i\hbar$ is not fundamental - it is the **Poisson bracket of the underlying geometric projections**, elevated to operator form in the legacy formalism.

Probability as Epistemic Ignorance. The Born rule ($P = |\psi|^2$) emerges not as a law of nature but as a **measure of phase coherence** in an incomplete network. If the global winding state were known, outcomes would be deterministic. Probability reflects **relational incompleteness**, not ontological randomness.

The End of the Measurement Problem. There is no “collapse.” There is only **phase locking** to satisfy energy symmetry with the environment. The observer is not special - they are part of the same phase network.

Theoretical Uroboros of Quantum Mechanics

Quantum mechanics is not a theory of particles or waves. It is the geometry of closed energy projections. Uncertainty, superposition, entanglement, and quantization are all necessary consequences of:

$$\boxed{\text{SPACETIME} \equiv \text{ENERGY}}$$

13 Unified Relational Geometry: Gravity versus Electromagnetism

Relational Consistency

The same **relational closure** applies at all scales:

$$\boxed{\kappa^2 = 2\beta^2} \longrightarrow \boxed{\kappa_q^2 = 2\beta_q^2}$$

Quantization is the discretization of geometric closure, not the introduction of a separate quantum postulate. Thus α ceases to be a ‘number about electromagnetism’ and becomes a geometric ratio defining the phase structure of all closed energy systems.

	Gravitational Case	Electromagnetic Case
Critical radius	$R_s = \frac{2GM}{c^2}$	$R_q = \frac{e_{\text{ch}}^2}{2\pi\epsilon_0 m_e c^2}$
Potential κ	$\kappa^2 = \frac{R_s}{r} = \frac{2GM}{c^2 r}$	$\kappa_q^2 = Z_e^2 \frac{R_q}{n^2 r_b}$
Orbital radius	$r = \frac{R_s}{\kappa^2}$	$r_n = \frac{Z_e R_q}{\kappa_q^2} = n^2 \frac{r_b}{Z_e}$
Kinetic β	$\beta^2 = \frac{R_s}{2r} \iff v = c\beta$	$\beta_q^2 = Z_e^2 \frac{R_q}{2n^2 r_b} \iff v_{EM} = c\beta_q$
Orbit Energy	$E_{\text{GR}} = \frac{m c^2}{2} \beta^2 = -\frac{GMm}{2r}$	$E_{\text{EM}} = \frac{m_e c^2}{2} \beta_q^2 = -\frac{m_e c^2 Z_e^2 \alpha^2}{2n^2}$

Both columns follow the *same algebraic geometry*:

$$\kappa^2 + \beta^2 = Q^2 \implies r = \frac{R_\bullet}{\kappa^2}, \quad E = \frac{m c^2}{2} \beta^2,$$

where $R_\bullet$ is R_s for gravity, R_q for electromagnetism, and Q^2 enforces the projectional balance.

The electromagnetic formulas above are identical in structure if one replaces $R_s \mapsto R_q$, $M \mapsto m_e$, $\kappa^2 \mapsto \kappa_q^2$, $\beta^2 \mapsto \beta_q^2$. Thus the same projection algebra yields both black-hole orbits and atomic orbits.

Summary

These results demonstrate that the WILL RG naturally unites gravitational and atomic physics, revealing a *single geometric origin* for phenomena traditionally treated by separate formalisms.

14 Why the Electron Does Not Collapse into the Nucleus: Topological and Ontological Resolution

14.1 Statement of the Problem

A longstanding paradox in both classical and early quantum theory is the apparent instability of the hydrogen atom: classically, an orbiting electron should continuously radiate energy and spiral into the nucleus, resulting in atomic collapse. However, atoms are empirically stable, and the electron remains at a finite distance from the nucleus in its ground state. Standard quantum mechanics resolves this paradox via the uncertainty principle and the existence of a lowest-energy stationary state. Here, we show that in the WILL geometric framework, atomic stability arises as a purely topological and ontological necessity, with no need for additional “quantum” postulates.

14.2 Geometric Condition for Stable Projection

Recall the geometric quantization condition:

$$n\lambda_n = 2\pi r_n, \quad n = 1, 2, 3, \dots \quad (19)$$

where n is the winding (topological) number, λ_n is the de Broglie wavelength, and r_n is the orbital radius for the n th energy level.

14.3 Ontological Meaning of $n = 0$

Within this framework, n has a strict ontological interpretation: it counts the number of complete phase rotations (windings) of the energy projection around the nucleus. The $n = 0$ case would correspond to zero phase winding—a state with no closed projection and thus no physical electron:

A system with $n = 0$ does not correspond to a physical electron bound to the nucleus; it represents the absence of any closed energetic projection. There is simply no object to “collapse.”

14.4 Mathematical Exclusion of Collapse

Suppose we attempt to reduce the orbital radius r_n to zero (i.e., electron “falling into the nucleus”). By the quantization condition:

$$r_n \rightarrow 0 \implies \lambda_n \rightarrow 0 \quad (20)$$

But from the de Broglie relation,

$$\lambda_n = \frac{h}{p_n} \implies p_n \rightarrow \infty \quad (21)$$

That is, the required electron momentum and energy diverge as $r_n \rightarrow 0$, making such a state energetically forbidden. Furthermore, the topological winding number n can only take integer values $n \geq 1$; $n = 0$ does not correspond to any physically realizable projection.

14.5 Minimal Stable State

Thus, the ground state ($n = 1$) is not just the lowest allowed energy configuration, but the minimal topologically permissible projection of energy around the nucleus:

$$n = 1 \iff \text{single closed winding of phase projection} \quad (22)$$

There is no valid state with $n < 1$; the electron cannot “collapse” further because the geometric structure of the system no longer exists.

14.6 Conclusion

The stability of the hydrogen atom, and the impossibility of the electron collapsing into the nucleus, arises naturally in the WILL framework as a consequence of topological closure. The electron’s existence as a bound system is identical to the existence of a nonzero winding number. There is no need to invoke additional quantum mechanical principles; the geometric ontology of energy projection alone guarantees atomic stability.

Summary

The electron exists only as a closed topological excitation; collapse would require its topology to vanish.

15 Hypothesis: Energy Symmetry as the Origin of Decoherence

Statement: Interference phenomena and coherent superpositions are only permitted in systems that are not energetically bound via interaction with **ANY** external object or system (including detector or measuring environment). The act of measurement corresponds to a physical interaction that invokes the principle of mutual energy conservation between the system and the detector:

$$\boxed{\Delta E_{A \rightarrow C} + \Delta E_{C \rightarrow A} = 0}$$

Interpretation: Before any interaction, the system's internal energy projection is unconstrained and may evolve or propagate through multiple coherent geometric trajectories simultaneously. Upon interaction, the requirement of energy symmetry forces the system to project into a single, well-defined energetic configuration. This projection eliminates the compatibility of multiple phase paths and thereby collapses the interference pattern.

Implications:

- Collapse is not epistemic (observer-dependent), but a geometric-energetic necessity arising from reciprocal closure.
- Decoherence is the energetic resolution of potential superposition into a single pathway dictated by energy-matching boundary conditions.
- Observation corresponds to a physical event, not a metaphysical concept.

Next Steps: To validate the hypothesis, we must:

- Test it against canonical quantum experiments (double slit, EPR, delayed choice).
- Model systems near the decoherence threshold (partial interaction).
- Explore energetic asymmetries between detector and system.

15.1 Empirical Validation: Decoherence from Pre-Interaction Events

Objective: To test the hypothesis that coherence and interference in quantum systems are disrupted not by epistemic acts of observation, but by physical energy exchange that enforces mutual energetic closure between the system and the environment.

Core Statement: A particle arriving at an interferometric structure (e.g., a double slit) with prior energetic entanglement (via scattering, emission, or thermal exchange) enters the system already constrained by symmetry:

$$\Delta E_{A \rightarrow C} + \Delta E_{C \rightarrow A} = 0$$

Therefore, its internal projection must resolve to a definite energetic configuration, preventing coherent interference.

15.2 Test Cases and Results

- **Case 1: Electrons in high vacuum** – Path length: 1 m; – Mean free path: 10–100 km; – Result: *Clear interference observed.* – **Conclusion:** No prior interaction $\Rightarrow$ full coherence preserved.
- **Case 2: C₆₀ molecules in ultra-high vacuum** – Coherent interference visible at low temperature; – When heated ($T > 3000\text{ K}$), thermal IR emission occurs; – *Interference pattern disappears.* – **Conclusion:** Internal energy leakage $\Rightarrow$ decoherence without measurement.
- **Case 3: Electrons in atmospheric air** – Mean free path $\sim 4\mu\text{m} \ll$ system size (0.1–1 m); – *No interference observed.* – **Conclusion:** High probability of pre-interaction $\Rightarrow$ loss of coherence.
- **Case 4: Photons with partial phase scattering before slit** – Partially diffusive medium inserted; – *Interference visibility decreases;* – **Conclusion:** Partial energy leakage $\Rightarrow$ partial loss of phase integrity.

15.3 Summary:

Across all tested regimes, the hypothesis holds:

- **Coherence exists** when no energetic connection exists between system and environment.
- **Interference disappears** when mutual energy conservation applies due to prior or current interactions.
- This behavior scales smoothly: partial interaction leads to partial decoherence, consistent with phase degradation models.

Implication: This supports a non-probabilistic, physically grounded account of quantum decoherence based purely on energy geometry, in full alignment with the foundational principle of WILL Geometry.

16 Addendum: The Geometric Resolution of Quantum Paradoxes

16.1 Relational Indeterminacy and the Mechanism of Collapse

Standard quantum mechanics posits probability as a fundamental feature of reality (Born rule). WILL Relational Geometry reinterprets quantum indeterminacy not as ontological randomness, but as **Relational Incompleteness**.

Definition 16.1 (Relational Indeterminacy). *Before a system interacts with an environment (detector), its internal phase projection ϕ is topologically defined (winding number n) but **relationally undefined** relative to the environment. The phase difference $\Delta\phi$ does not exist as a fixed value until a causal channel is established. This state of "floating phase" allows for the coexistence of multiple geometric trajectories (superposition).*

Principle 16.2 (Collapse via Energy Symmetry). *"Measurement" is the physical enforcement of the **Energy Symmetry Law** (result from WILL RG Part I):*

$$\Delta E_{sys \rightarrow env} + \Delta E_{env \rightarrow sys} = 0$$

Upon interaction, the system is forced to adopt a specific phase configuration because only discrete phase values satisfy the strict algebraic requirement of energy conservation. "Collapse" is the deterministic solution to this boundary value problem.

16.2 Entanglement as Topological Unity

Entanglement is re-defined not as a signal between two particles, but as the property of a **Single Energy System** projected onto spatially separated coordinates.

Theorem 16.3 (Systemic Integrity). *Two entangled particles A and B are not distinct entities but projections of a single phase network S_{AB} .*

$$\Psi_{AB} \neq \psi_A \otimes \psi_B$$

*The "connection" between A and B is a topological constraint of the vacuum geometry that holds the system together. While massive particles possess proper time ($d\tau > 0$), their **correlation** is a property of the phase network, which resides in the geometric structure of the vacuum and is not subject to speed limits. Interaction with A constrains the global topology of S_{AB} , instantaneously resolving the state of B not via superluminal signaling, but via the definition of the system itself.*

16.3 Resolution of the "Delayed Choice" Paradox (Quantum Eraser)

The paradox of future measurements influencing past interference patterns (Kim et al., 1999) is resolved by applying the geometry of the null interval strictly.

Proposition 16.4 (Null Interval Simultaneity). *For a photon ($\beta = 1$), the phase projection is zero ($\beta_Y = 0$). Consequently, the spacetime interval along its path is null:*

$$ds^2 = 0 \quad \text{and} \quad d\tau = 0.$$

Resolution: In the reference frame of the interaction carrier (light), the emission event, the mirrors, and the detectors exist at a **single point of causal contact**. There is no "flight time." The configuration of the detectors (even if modified "later" in the laboratory frame) acts as a **boundary condition** of the single topological network at the "moment" of emission. The photon does not predict the future; it occupies a geometry where the emitter and detector are adjacent. Retrocausality is an illusion caused by imposing a massive observer's time perception ($\beta_Y \approx 1$) onto a massless entity ($\beta_Y = 0$).

16.4 Resolution of Interaction-Free Measurement (Bomb Tester)

The paradox of detecting an object without energy exchange (Elitzur-Vaidman) is resolved by analyzing the topology of the phase network.

Resolution: The presence of the "bomb" changes the global topology of the interferometer from a **Closed Loop** (where destructive interference forbids energy flow to the dark detector) to an **Open Fork**. When a photon is detected at the dark detector (D_{dark}),

a real energy exchange occurs: $\Delta E_{sys \rightarrow D_{dark}} \neq 0$. This event is geometrically permissible *only* within the "Fork" topology. The system detects the bomb not by "touching" it with the photon, but because the bomb's presence altered the **geometry of the phase space**, enabling an energy transfer at D_{dark} that was previously forbidden by topological closure.

17 Conclusion

By a rigorous application of the principles of WILL Relational Geometry, we have derived the complete structure of the hydrogen atom. We have demonstrated that quantization, the Bohr radius, the fine structure constant, and the discrete energy spectrum are not independent physical postulates but are all necessary and inevitable consequences of a single, unified principle: that reality is described by a closed, self-consistent relational geometry.

The framework requires no classical force analogues, no probabilistic axioms, and no external postulates. Atomic structure is shown to be a direct manifestation of the geometry of energy itself. What physics calls 'quantum mechanics' is the algebraic projection of this geometric unity.